

ReadSpyn: A Semi-Classical Simulator for RF Reflectometry Readout of Quantum Dot Systems

Model Summary, Physical Assumptions, and
Path Towards Realistic Simulation

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Abstract

We present READSPYN, a time-domain simulator for RF reflectometry readout of semiconductor quantum dot (QD) systems. The simulator propagates the equations of motion of an RLC resonator capacitively coupled to one or more quantum dots, driven at its resonant frequency, and subject to charge noise. The reflected voltage is subsequently IQ-demodulated to yield the quadrature signals $I(t)$ and $Q(t)$ recorded in experiment. READSPYN represents a step towards realistically reproducing laboratory readout traces: it incorporates an experimentally motivated charge-noise spectrum, an analytic Coulomb-peak conductance model, and a self-consistent circuit ODE. We explicitly enumerate the physical approximations made and identify the extensions required to close the gap with state-of-the-art experimental conditions.

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1 Introduction and Motivation

Radio-frequency (RF) reflectometry is the workhorse readout technique for spin and charge qubits in semiconductor QD devices [1, 2, 3]. A resonant circuit is tank-matched to a sensor QD; changes in the sensor conductance shift the resonator’s impedance and are imprinted on the phase and amplitude of the reflected microwave tone. After IQ demodulation the two quadratures $I(t)$ and $Q(t)$ are used to discriminate charge states in real time.

Simulating this measurement chain end-to-end is valuable because (i) it allows optimisation of device parameters before fabrication, (ii) it provides ground-truth data for benchmarking machine-learning classifiers, and (iii) it enables systematic studies of how different noise sources limit readout fidelity.

READSPYN targets these use cases within a semi-classical framework that is tractable and interpretable while capturing the dominant physics. This document summarises the model, states the assumptions explicitly, and outlines the road map towards higher-fidelity simulation.

2 Quantum Dot System

2.1 Constant-Interaction Model

The electrostatics of an array of N quantum dots are described within the constant-interaction (CI) model. Mutual capacitances between dots and between dots and sensors are encoded in the matrices $C_{dd} \in \mathbb{R}^{N \times N}$ and $C_{ds} \in \mathbb{R}^{N \times M}$, where M is the number of sensor QDs.

2.2 Effective Detuning

Given a charge-state vector $\mathbf{n} \in \mathbb{Z}^N$, a sensor-gate voltage vector $\mathbf{V}_s \in \mathbb{R}^M$, and a global offset ε_0 , the effective detuning of sensor j is

$$\varepsilon_j = [C_{ds}^\top C_{dd}^{-1}]_j (\mathbf{n} + C_{ds} \mathbf{V}_s) + \varepsilon_0. \quad (1)$$

The row vector $[C_{ds}^\top C_{dd}^{-1}]_j$ is the *lever arm* projecting the charge configuration onto the detuning axis of sensor j . Different charge states $\mathbf{n}^{(A)}$ and $\mathbf{n}^{(B)}$ produce different detunings $\varepsilon_j^{(A)}$ and $\varepsilon_j^{(B)}$, which is the physical origin of the IQ contrast used for charge-state discrimination.

3 Sensor Conductance: the Coulomb Peak

The sensor QD is operated near a Coulomb blockade resonance. In the sequential-tunnelling limit at low temperature the zero-bias conductance as a function of detuning is [4]

$$G(\varepsilon) = \frac{2}{R_0} \cosh^{-2} \left(\frac{2\varepsilon}{\varepsilon_w} \right), \quad (2)$$

where $R_0 = 1/g_0$ is the on-resonance resistance and ε_w is the thermal broadening width. The local slope

$$\frac{dG}{d\varepsilon} = -\frac{4}{R_0 \varepsilon_w} \cosh^{-2} \left(\frac{2\varepsilon}{\varepsilon_w} \right) \tanh \left(\frac{2\varepsilon}{\varepsilon_w} \right) \quad (3)$$

is zero at the peak centre and extremal at $|\varepsilon| = \varepsilon_w/2$. A charge state whose static detuning (1) lands on the steep flank therefore produces larger IQ fluctuations under charge noise than one sitting near the peak top — even when both states experience the same noise. This is the mechanism behind the asymmetric noise sensitivity observed in the two-dot simulations.

4 RLC Resonator and Circuit ODE

4.1 Lumped-Element Circuit

The sensor QD is embedded in a lumped RLC tank circuit:

- Inductor L_c (bond wire or on-chip spiral),
- Parasitic shunt capacitance C_p ,
- Transmission-line load R_L (typically $50\ \Omega$),
- Optional series coupling resistor R_c .

The circuit is driven at $\omega_d \approx \omega_0 = 1/\sqrt{L_c C_p}$.

4.2 Equations of Motion

With state vector $\mathbf{y} = (V_{C_p}, I_L)^\top$ the circuit obeys

$$\frac{dV_{C_p}}{dt} = \frac{I_L - G_s(t)V_{C_p}}{C_p} \quad (R_c = 0), \quad (4)$$

$$\frac{dI_L}{dt} = \frac{V_s(t) - R_L I_L - V_A}{L_c}, \quad (5)$$

where $V_s(t) = V_0 \cos(\omega_d t)$ and $V_A = V_{C_p}$ for $R_c = 0$. The time-dependent conductance $G_s(t) = G(\varepsilon(t))$ is obtained by substituting the instantaneous noisy detuning into (2). The ODEs are integrated with the stiff Radau solver (`scipy.integrate.solve_ivp`); inner-loop functions are JIT-compiled with Numba.

4.3 IQ Demodulation

The voltage $V_A(t)$ is demodulated against the drive tone:

$$I(t) = \text{LPF}[V_A(t) \cos(\omega_d t + \phi)], \quad (6)$$

$$Q(t) = \text{LPF}[V_A(t) \sin(\omega_d t + \phi)], \quad (7)$$

where ϕ is the demodulation phase. In READSPYN demodulation uses the Hilbert transform; edge artefacts are suppressed by padding the integration window before trimming to the requested output interval.

5 Noise Models

5.1 Spectrum-Prescribed $1/f$ Noise

The dominant low-frequency noise in semiconductor QD devices is charge noise with PSD $S(f) \propto 1/f$. READSPYN generates noise trajectories $\delta\varepsilon(t)$ by drawing Gaussian random amplitudes in frequency space scaled by $\sqrt{S(f)}$ and inverse-Fourier-transforming, then normalising to a prescribed RMS amplitude σ (in units of ε_w). The same realisation can be shared across charge states (common-mode noise) or independent realisations can be used.

5.2 Ornstein–Uhlenbeck and Telegraph Noise

For coloured Gaussian noise with a Lorentzian PSD, an Ornstein–Uhlenbeck process is available:

$$d\delta\varepsilon = -\gamma \delta\varepsilon dt + \sigma \sqrt{2\gamma} dW, \quad (8)$$

with correlation time $\tau_c = 1/\gamma$. Two-level fluctuators (bistable charge traps) are modelled as telegraph noise with switching rate γ .

6 A Step Towards Realistic Simulation

READSPYN advances beyond a simple analytic toy model in several respects:

1. **Self-consistent circuit dynamics.** The conductance back-action on the resonator is solved via a full ODE rather than a perturbative susceptibility formula.
2. **Time-domain noise injection.** Noise enters as a time-domain trajectory, so simulated IQ traces include realistic low-frequency drift.
3. **User-configurable spectral shape.** The $1/f$ PSD can be matched to measured charge-noise spectra.
4. **Multi-dot, multi-sensor geometry.** The capacitance-matrix formalism supports arbitrary numbers of dots and sensors, enabling studies of cross-talk and lever-arm asymmetries.
5. **Edge-corrected Hilbert demodulation.** Window padding suppresses Hilbert-transform edge ringing, mirroring experimental DSP practice.
6. **Stationary initial state.** The ODE is initialised at the ring-up-free steady state, avoiding transient artefacts.

7 Limitations with Respect to Realistic Semiconductor QD Measurements

Despite the advances above, several important physical effects are absent from the current framework. We group them thematically below.

7.1 Quantum and Many-Body Effects

- a. **No tunnel broadening.** Eq. (2) is the sequential-tunnelling result valid at $k_B T \gg \Gamma$. When $\Gamma \sim k_B T$ the Lorentzian (co-tunnelling / Kondo) line shape is not captured.
- b. **Classical charge states only.** Coherent charge superpositions (e.g. spin-to-charge conversion in Pauli-spin-blockade readout) require a density-matrix description.
- c. **No spin, valley, or orbital degeneracy.** Zeeman splitting, spin-orbit coupling, and valley degeneracy in Si/Ge are absent.
- d. **Constant-interaction approximation.** Energy-dependent screening and higher-order Coulomb effects are neglected.

7.2 Thermal and Non-Equilibrium Effects

- a. **Fixed electron temperature.** Microwave heating of the electron bath (photon-assisted tunnelling) is not modelled.
- b. **No photon-number-dependent back-action.** Above the single-photon power limit the resonator photon number causes an ac-Stark shift and measurement-induced dephasing, absent in the classical ODE.
- c. **No charge-state relaxation.** The charge configuration $\mathbf{n}$ is held constant during integration; qubit T_1 processes are not simulated.

7.3 Circuit and Electromagnetic Effects

- a. **Lumped-element model.** Distributed transmission-line effects and parasitic resonances of bond wires become important above ~ 1 GHz.
- b. **No amplifier noise.** The noise floor in experiment is dominated by the cryogenic amplifier (HEMT or paramp); this is not added to the simulated signal.
- c. **Perfect impedance matching.** Reflections from mismatch between the tank and the coaxial line are ignored.
- d. **No dispersive frequency shift.** Only the dissipative part δG_s is modelled; the reactive part δB_s that shifts the resonant frequency (dispersive readout) is absent.

7.4 Noise Modelling

- a. **Scalar, additive, global charge noise.** The noise is a single shift $\delta\varepsilon(t)$ on the sensor detuning. Per-gate voltage fluctuations that perturb individual dot potentials independently are not resolved.
- b. **No spatial cross-correlations.** Correlated noise from a single nearby charge trap affecting multiple dots is not included.
- c. **No TLS noise in circuit components.** Two-level-system noise in the dielectric of C_p causes resistance fluctuations that are not modelled.

7.5 Signal Processing

- a. **Ideal local oscillator.** Phase noise and frequency jitter of the drive source are absent.
- b. **Ideal brick-wall low-pass filter.** The Hilbert-transform demodulation acts as a brick-wall filter; a realistic Butterworth or Gaussian filter would roll off differently.
- c. **No ADC effects.** Quantisation noise, dynamic range limitations, and clock jitter are not included.

8 Road Map

Table 1 summarises the most impactful extensions and their estimated implementation effort.

Extension	Physical motivation	Effort
Amplifier noise (HEMT / paramp)	Sets SNR floor in all experiments	Low
Finite electron temperature T_e	Low-power / high-fidelity regime	Low
Per-dot local gate noise	Realistic multi-dot operation	Low
Charge relaxation / T_1 during readout	Spin-qubit readout fidelity	Medium
Dispersive frequency shift (δB_s)	Dispersive readout schemes	Medium
Tunnel-broadened Lorentzian line shape	$\Gamma \sim k_B T$ regime	Medium
Density-matrix charge dynamics	Coherent charge superpositions	High
Distributed circuit / transmission line	$f_0 > 1$ GHz, parasitic resonances	High

Table 1: Prioritised road map for extending READSPYN.

9 Conclusion

READSPYN provides a self-consistent, time-domain simulation of RF reflectometry readout of semiconductor quantum dot systems. By solving the circuit ODE in the presence of physically motivated charge noise and demodulating the output, the simulator reproduces the qualitative features of experimental IQ traces: state-dependent DC offsets, noise-driven fluctuations whose magnitude depends on the local slope (3) of the Coulomb peak, and comet-tail trajectories in the IQ plane.

The framework is a deliberate step towards realistic measurement simulation rather than a full first-principles model. Section 7 enumerates the current approximations in order of physical importance. The most pressing gaps are the absence of amplifier noise (which sets the actual SNR floor) and of charge-relaxation dynamics (which limit spin-qubit readout fidelity). Addressing these two items, together with per-dot local gate noise for multi-dot arrays, would bring READSPYN into quantitative contact with state-of-the-art experimental data.

References

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